Input Data
300 curated donor-acceptor device records
Device labels: Jsc, Voc, FF, PCE, energy loss

Molecular Standardization
RDKit parsing, sanitization, and canonicalization
Separate donor and acceptor populations

Hybrid Molecular Generation
12 proposal and augmentation streams
Neural, graph, fragment, mutation, and recombination routes
Donor pool
Acceptor pool

Chemical Validity and Novelty Filtering
Sanitization and single-component checks
Exact-match novelty against training set
Duplicate removal
Adaptive Tanimoto near-neighbor novelty filter

Rejected structures

Synthesizability Assessment
SAScore and RAScore screening
Building-block compatibility scoring

Prioritization attribute, not a hard gate

Donor-Acceptor Pairing
Trait-balanced Cartesian pairing
11,988,459 combinations

Property Prediction and Prioritization
Ridge regression surrogate with Gaussian-process correction
Predictions for PCE, Jsc, Voc, FF, and energy loss
Tiered shortlisting with structure-disjoint validation

Prioritized Candidates
52,995 donor candidates and 53,719 acceptor candidates
2,832 Tier-2 donor-acceptor pairs for experimental follow-up